

Final Year Project Proposal

Topic

Blockchain and ML integration in
scholarship management

Submitted By

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Title:"Blockchain and Machine Learning Integration in Scholarship Management"

Abstract:

Scholarship management systems frequently encounter challenges such as inefficiencies, lack of transparency, and serious data privacy concerns. These issues obstruct the fair distribution of educational resources and complicate the scholarship allocation process. This research proposes a blockchain-based platform to mitigate these challenges. While blockchain technology offers numerous advantages, it also introduces scalability issues that can be addressed through zk-rollup solutions. The integration of Self-Sovereign Identity (SSI) and Zero-Knowledge Proof (ZKP) technologies ensures secure and private submission of scholarship applications while maintaining transparency in the verification process. Additionally, a machine learning model is developed to assess scholarship eligibility, acting as a recommendation system that identifies and prioritizes qualified candidates. This data-driven approach aims to remove barriers that may prevent deserving students from accessing financial aid, such as administrative errors or misconceptions about eligibility. Experimental results indicate that zk-rollups enhance transaction efficiency, reducing confirmation times by approximately 63.6% and transaction costs by nearly 90%. Furthermore, the machine learning model demonstrated strong performance, achieving a balanced accuracy of 86.75% and a mean average precision of 91.68% on a realistically imbalanced dataset.

1. Introduction/Background

Scholarship management systems are pivotal in providing financial assistance to deserving students, yet many traditional systems face significant challenges. These include inefficiencies in fund allocation, lack of transparency, and concerns over data privacy. Centralized databases often lead to errors in fund distribution, mistrust, and administrative overload. In response to these challenges, this project proposes an innovative solution that combines blockchain technology with machine learning (ML) to create a transparent, secure, and efficient scholarship management platform. The integration of Zero-Knowledge Proofs (ZKPs) and zk-rollups ensures both privacy and scalability, addressing the limitations of existing systems.

2. Objectives

The primary objectives of this project are:

To evaluate the effectiveness of integrating blockchain with machine learning in the scholarship management process.

To enhance transparency and reduce inefficiencies in scholarship allocation through decentralized blockchain technology.

To develop a machine learning model that predicts scholarship eligibility based on multiple factors (academic performance, financial need, etc.).

To integrate zk-rollups for scalability and Zero-Knowledge Proofs (ZKPs) to ensure the privacy of applicants' sensitive data.

To analyze the impact of these technologies on the overall efficiency, cost, and fairness of scholarship distribution.

3. Methodology

The project will adopt a hybrid approach combining both primary data collection and technology development:

Blockchain Integration: The use of blockchain technology will ensure that all scholarship-related data (applications, eligibility checks, fund distribution) is stored transparently on a decentralized ledger.

Machine Learning: A Random Forest model will be trained using historical scholarship application data to predict eligibility. This model will consider features such as GPA, family income, extracurricular activities, etc.

zk-Rollups: These will be used to improve blockchain scalability, bundling multiple transactions into one proof to reduce transaction costs and improve processing speed.

Zero-Knowledge Proofs (ZKPs): ZKPs will be integrated to allow verification of a student's eligibility without exposing sensitive personal data.

The methodology will involve:

1. Collecting data from scholarship applications (including academic and financial data).
2. Developing and training the ML model for eligibility prediction.
3. Integrating blockchain for secure data storage and transaction management.
4. Implementing zk-rollups and ZKPs for scalability and privacy.
5. Testing the system's performance in real-world scenarios.

4. Scope and Delimitations

Scope: The project will focus on designing and testing a blockchain-based scholarship management platform for academic scholarships, utilizing machine learning for eligibility predictions and zk-rollups for scalability.

Delimitations: The scope of the project is limited to:

- oUrban areas with a significant population of scholarship applicants (100,000+).
- oOnly academic scholarships (excluding sports or merit-based scholarships).
- oThe project will not address legal, regulatory, or ethical issues related to data privacy laws like GDPR but will focus on technology implementation.

5. Expected Outcomes :The expected outcomes of the project include:

A functional prototype of a blockchain and ML-powered scholarship management platform.

Enhanced transparency in scholarship allocation, allowing all stakeholders to audit processes.

Improved efficiency, with reduced transaction times and administrative overhead, made possible by zk-rollups.

A machine learning model that can predict scholarship eligibility with high accuracy (expected to achieve a balanced accuracy of over 85%).

A comprehensive analysis of cost and time savings, showcasing the advantages of the integrated system over traditional scholarship management processes.

6. Significance of the Project

This project will have significant implications for scholarship management systems, particularly in addressing common issues of inefficiency, data privacy, and transparency. By integrating blockchain with machine learning, the project will:

Improve the fairness of scholarship distribution by ensuring that the eligibility process is data-driven and transparent.

Ensure security and privacy for sensitive applicant data through blockchain encryption and Zero-Knowledge Proofs.

Facilitate scalability through zk-rollups, which will allow the system to manage a large number of applications without compromising on performance.

Provide a model for future digital scholarship systems, influencing both policy and technological decisions at academic institutions and governmental bodies.

7. Timeline

Month 1-2:

- oConduct a literature review on blockchain applications in scholarship management and the integration of machine learning for predictive analytics.

- oData collection: Gather historical scholarship application data (academic performance, financial background, extracurriculars).

Month 3-4:

- oDevelop blockchain architecture for the scholarship management platform.

- oTrain the machine learning model (Random Forest) to predict scholarship eligibility.

- oImplement zk-rollups and Zero-Knowledge Proofs for scalability and privacy.

Month 5:

- oTesting and Evaluation: Test the system's performance, including transaction efficiency, cost, and model accuracy.

- oFinalize project report with results, conclusions, and recommendations for future research.

8. Resources and Budget

Resources Needed:

- oHuman: Blockchain developers, machine learning specialists, data analysts.

- oMaterial: Access to historical scholarship application data, computing resources for machine learning model training.

- oSoftware: Blockchain platform (e.g., Ethereum), machine learning tools (Python, Scikit-learn, TensorFlow), zk-rollup and ZKP integration tools.

Budget Estimate:

- oData collection and analysis tools: 1,000

- oBlockchain platform setup and hosting: 1,500

- oML model development (software licenses, tools): 1,000

- oMiscellaneous (testing and evaluation): 500

Total Budget: 4,000

9. Conclusion

This project aims to address critical challenges in scholarship management, such as inefficiency, transparency, and data privacy concerns, through the innovative use of blockchain and machine learning technologies. By integrating zk-rollups for scalability and Zero-Knowledge Proofs for privacy, the project seeks to enhance both the transparency and security of scholarship allocation processes. The results of this study will provide valuable insights into how modern technologies can improve administrative processes, ensuring that scholarships are awarded fairly and efficiently, expanding access to educational opportunities for all students.